

Serum enterolactone levels and mortality outcome in women with early breast cancer: a retrospective cohort study.

We previously demonstrated that high serum enterolactone levels are associated with a reduced incidence of breast cancer in healthy women. The present study was aimed at investigating whether a similar association might be found between serum enterolactone levels and the mortality of women with early breast cancer. The levels of enterolactone in cryopreserved serum aliquots obtained from 300 patients, operated on for breast cancer, were measured using a time-resolved fluoro-immunoassay. Levels were analyzed in respect to the risk of mortality following surgery. Cox proportional hazard regression models were used to check for prognostic features, to estimate hazard ratios for group comparisons and to test for the interaction on mortality hazards between the variables and enterolactone concentrations. The Fine and Gray competing risk proportional hazard regression model was used to predict the probabilities of breast cancer-related and breast cancer-unrelated mortalities. At a median follow-up time of 23 years (range 0.6-26.1), 180 patients died, 112 of whom died due to breast cancer-related events. An association between a decreased mortality risk and enterolactone levels ≥ 10 nmol/l was found in respect to both all-cause and breast cancer-specific mortality. The difference in mortality hazards was statistically significant, but it appeared to decrease and to lose significance after the first 10 years, though competing risk analysis showed that breast cancer-related mortality risk remained constantly lower in those patients with higher enterolactone levels. Our findings are consistent with those of most recent literature and provide further evidence that mammalian lignans might play an important role in reducing all-cause and cancer-specific mortality of the patients operated on for breast cancer.